

USDA License No. XXX

BP MFG US 2000L SUB.xx

Batch # _____

USDA Code: XXXX.XX

PO # _____

BLDG/RM # _____ / _____

**Production Records for Sample Virus in 2000L Single Use Bioreactor (SUB) For
U.S. Use**

Procedure	Data Entries
I. Virus Seed Identification Master Seed Identification: _____ Last Testing Date: ____ / ____ / ____ Max Passage of seed is X+6	Working Seed Identification: Passage: _____ X+ _____ Batch #1: _____ Volume: _____ mL Titer: _____ Batch #2: _____ Volume: _____ mL Titer: _____ Batch #3: _____ Volume: _____ mL Titer: _____
II. Cell Identification: Master Cell Stock Batch# _____ - _____ Last Testing date ____ / ____ / ____ Max passage of cells must not exceed X+30	Production Stock Cell Identification #: _____ Batch #: _____ Batch Date: _____ Cell Bioreactor #: _____ By: _____ Check By: _____
III. Production Vessel Preparation (Refer to SOP XX-XXX-X to calibrate probes)	Date: _____

Procedure	Data Entries
A. Electrochemical pH probes calibration.	Date: _____ By: _____
Attach probes sterilization sticker on MD	By: _____
Follow SOP XX-XXX-X to install the single use bag in the SUB.	
NOTE: Close 2 of the 4 SUB exhaust filters to ensure that the bag does not accumulate pressure. SUB is not designed for any pressure load. Also verify that the vent filters are not kinked.	By: _____
B. Vessel Identification Vessel Number: _____ Vessel Type: <u>2000L SUB</u>	Single Use Bag Lot #: _____ Expiration Date: _____ By: _____ Check By: _____
C. Probe Identification	DO Probe No.: _____ Sleeve Serial No.: _____ DO Probe No.: _____ Sleeve Serial No.: _____ TruFluor DO Probe No.: _____ Sleeve Serial No.: _____ pH Probe No.: _____ Sleeve Serial No.: _____ TruFluor pH Probe No.: _____ Sleeve Serial No.: _____

Procedure	Data Entries
D. Clamp off all lines to bag except for exhaust, overlay, the hollow spargers and probe lines.	By: _____ Check By: _____
E. Connect overlay (Air Flow-1) and pressure sensor to the controller.	By: _____ Check By: _____
F. Connect sparger 1 from the controller to the hollow sparger at bottom of the bag.	By: _____ Check By: _____
G. Insert DO and pH probes into bag and connect their respective cables.	By: _____ Check By: _____
H. Insert temperature probe into bag.	By: _____ Check By: _____
IV. Procedures for Media Preparation	
NOTE: Refer to SOP XX-XXX-X to operate the controller.	
A. From the controller's main screen, verify that the gases are flowing into the right port: Air Flow-1 to overlay and Air Flow-2, N2, CO2, O2 Flow-1, O2 Flow-2 to the hollow sparger.	By: _____ Check By: _____
B. Visually inspect the bottom of the bag for wrinkles. Open the media fill line before transferring growth media to reactor and record the amount in the Component Table.	By: _____ Check By: _____

Procedure	Data Entries
C. Calculate volume to fill reactor to before seed addition.	Planned final Volume: _____ Kg Seed Volume: - _____ Kg Fil Volume: = _____ Kg Calculations By: _____ Check By: _____
Note: Growth Media may be transferred before inoculation day; however, all the other components must be added the day of infection.	

Standard Volume: 2000L

Inoculation Component Table:

Components	Manufacturer	Amount	Batch Number	Exp Date	Amount Added in L	By	Check By	SAP
Excell	In House	QS to 2000L			L			
Insect Cell Suspension	In House	1.0x10 ⁶ Cells/mL		N/A	L			
200 mM L-Glutamine*	In House	20mL/L			L			
Gentamycin*	In House	0.1mL/L			L			
Antifoam*	In House	<0.2 mL/L			L			

*Amounts calculated per planned final volume.

D. Turn on the TCU connected to the SUB and set the following Setpoints on the controller according to SOP XX-XXX-X. Temperature: 2x°C Agitator: 5x RPM Vent Heater 1& 2: 3x°C	Date: _____ Time: _____ AM / PM By: _____ Check By: _____
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Procedure	Data Entries
E. Calibrate the electro-chemical DO probes. Once the vessel has reached the temperature set point, set the 0% and the 100% calibration points. (Both probes must be calibrated) Refer to SOP XX-XXX-X to calibrate DO probes and enter the calibration values.	DO Probe #1 Zero/: _____ nA Slope (span): _____ nA DO Probe #2 Zero/: _____ nA Slope (span): _____ nA By: _____
1. Set the DO control to xx% and put it on "Auto". Reduce stirrer back to xx RPM. 2. Zero O₂ totalizer.	Date: _____ Time: _____ AM / PM By: _____ Check By: _____
F. LAF Sanitization Disinfectant used: <u>X% H₂O₂</u>	Room #: _____ LAF #: _____ Batch #: _____ Exp. Date: _____ By: _____
1. Aseptically, in a sterile vessel, add the appropriate amount of xxxmM L-Glutamine, Gentamycin, and antifoam according to the component table.	By: _____ Check By: _____
G. Calibrate pH probe. Calibrate a bench-top pH meter per SOP XX-XXX. Adjust Bioreactor pH probes offset so they match the offline reading. Verify that the pH controller is in manual mode, no need to monitor pH for this process.	By: _____ Check By: _____ Final pH at Plant: _____

Procedure	Data Entries
V. Procedures for Plant / Infection	
A. Receive cells from the cell prep lab.	Date Cells Received: _____ Time Cells Received: _____AM / PM By: _____
B. Attach cell prep information to this MD.	By: _____
C. Ensure the media is at 2x°C before transferring cells to the reactor. If using a tank, up to 10 pounds of air pressure may be applied to it to transfer cells into the reactor.	Cell Tank #: _____ Sterilization Exp. Date: _____ Time: _____AM / PM By: _____ Check By: _____
D. Upon reaching Fill Volume (from step IV.C.), let the cells mix for approximately 5 minutes in the SUB, then collect a sample and fill out the Daily Observations table. When the SUB has reached a cell density of approximately 1.0×10^6 cells/mL, aseptically add thawed seed.	Cell Count: _____x 10^6 cells/mL Start Time: _____AM / PM Finish Time: _____AM / PM By: _____
E. If multiple batches of seed are used, calculate a weighted average with the following chart. Refer to Step I.	<u>Seed Batch #1:</u> Inv Log of Titer: _____ Seed Volume: X _____mL Weighted Log: = _____ <u>Seed Batch #2:</u> Inv Log of Titer: _____ Seed Volume: X _____mL Weighted Log: = _____

Procedure	Data Entries
	Seed Batch #3: Inv Log of Titer: _____ Seed Volume: X _____mL Weighted Log: = _____ Total Weighted Log: _____ Total Volume ÷ _____mL Final Inv. Log of Titer = _____
F. Calculate amount of virus seed needed using planned conditions for infection: Reactor Fill Volume = reactor volume before seed added (from step IV.C.) Cell Density = 1.0×10^6 cells/mL (standard cell density for infection) 0.2x = standard MOI for infection in bioreactor Inv. Log Titer = $\log^{-1}$ of seed titer (from step V.F. if multiple batches used)	Reactor Fill Volume: _____mL Cell Density: X <u>1.0×10^6 cells/mL</u> MOI: X <u>0.2x</u> Inv. Log Titer: ÷ _____ Amount Seed Needed = _____mL Calculations By: _____ Check By: _____
G. Thaw the amount of virus seed required in a bath of disinfectant. The bath should be no warmer than $2x^\circ\text{C}$. Add the thawed seed to Bioreactor.	By: _____ Check By: _____
H. Calculate Actual MOI as follows: Actual amount of seed = volume of seed added to reactor Inv. Log Titer = $\log^{-1}$ of seed titer (from step V.F. if multiple batches used) Cell Density = cell count at time of infection (from step V.E.) Reactor Fill Volume = reactor volume before seed added (from step IV.C.)	Actual Amount Seed Used _____mL Inv. Log Titer: X _____ Cell Density: ÷ _____ Reactor Fill Volume ÷ _____mL Actual Plant MOI = _____ Calculations By: _____ Check By: _____

Procedure	Data Entries
I. Clipster off the lines used to transfer components, cells, and seed to the reactor.	By: _____ Check By: _____
J. On the configuration screen of the controller go to "Vessel settings management" from the scroll down menu, select and load the "XXX_Alarms_File" file. Label the file using this format: Batch #_Alarms_ON .	By: _____ Check By: _____
K. Record Inoculation finish time and label the SUB as follows: Virus code / serial # / date.	Time: _____AM / PM By: _____ Check By: _____
L. Any remaining cells and seed must be killed with Hypochlor (as per VSM 800.56) at 100mL/L and discarded. A minimum of 10 minutes of disinfectant contact time is required before discard.	Volume of Seed Discarded: _____mL Volume of Cells Discarded: _____mL Disinfectant Batch #: _____ Exp. Date: _____ Start Time: _____AM / PM End Time: _____AM / PM By: _____ Check By: _____
M. The reactor is sampled daily and results are recorded in the daily observations table. NOTE: When sampling, ensure the following parameters are in range. If out of range, contact Supervisor and comment on any corrective action taken. Temperature: 2x°C Dissolved oxygen: xx%	

Procedure		Data Entries				
DAILY OBSERVATIONS						
TEMP °C (25-29°C)	°C	°C	°C	°C	°C	°C
RPM (55)	rpm	rpm	rpm	rpm	rpm	rpm
pH / Control Probe #	/ #	/ #	/ #	/ #	/ #	/ #
External pH						
DO% / Control Probe (20-100%)	/ #	/ #	/ #	/ #	/ #	/ #
O ₂ Consumption	L	L	L	L	L	L
Visual (Microscopic) CPE%						
Cell count	Live= Dead=	Live= Dead=	Live= Dead=	Live= Dead=	Live= Dead=	Live= Dead=
Time	AM / PM	AM / PM	AM / PM	AM / PM	AM / PM	AM / PM
Date						
By						
Note: DO% may rise above 100% close to harvest time.						
VI. Harvest of Antigen Note: Unifuge must be connected to the harvest port of the SUB bag. Note: See SOP (XX-XXX-X) Operation of the Unifuge Centrifuge when setting up for harvest. Note: Discard waste stream will include a sterile carboy with station connected aseptically to the Unifuge Supernatant line. The sterile carboy will then be connected to a discard vessel. Follow the same procedures for both Unifuge if using two.				Date: _____ Unifuge 1 #: _____ Unifuge 2 #: _____ Module 1 Lot / S/N #: _____ Module 2 Lot / S/N #: _____ Harvest Tank #: _____ Sterilization Exp. Date: _____ By: _____ Check By: _____		

Procedure	Data Entries
A. Before harvest, go to the configuration screen, load the “Vessel_Process_Alarms_Off” .	By: _____ Check By: _____
B. Begin harvest at ~XXX hr after time of infection.	SUB Volume at Start of Harvest: _____ L By: _____
C. As described in SOP XX-XXX-X, verify that the feed tube is welded to the harvest line of the SUB, supernatant tube to discard vessel, cell concentrate tube to harvest tank, and rinse tube to PBS.	
D. Tare the harvest tank on the floor scale before harvesting.	Time Transfer Start: _____ AM/PM By: _____ Check By: _____
E. Harvest the antigen by passing 100% of the bioreactor volume through Unifuge. Centrifuge at a target rate of 3 liters per minute at 3000G's.	
F. After harvesting approximately 60% of the SUB volume (~1200L for a 2000L batch), turn off vent heaters, agitator, DO, and temperature control.	By: _____ Check By: _____
G. Record the PBS amount used during harvest.	PBS Batch #: _____ Amount Added: _____ L SAP Write off _____ L Exp. Date: _____ By: _____ Check By: _____

Procedure	Data Entries
H. Continue the harvest until 100% of the bioreactor volume has passed through Unifuge. - Harvest tank will be filled to ~10% of bioreactor volume (ex. 2000kg bioreactor volume will yield 200kg harvest) - Record the weight of the Harvest tank after harvest has finished.	Harvest Tank Weight: _____ kg Time: _____ AM / PM By: _____ Check By: _____
I. Air pressure or peristaltic pump may be used to transfer antigen from the harvest tank into carboys. If using air pressure, properly vent the air filters from the carboy stations to avoid pressure build-up in the carboys once in the freezer. Before aseptically disconnecting the carboy filters, verify that the pressure gauge on the harvest tank is reading 0 psi.	By: _____ Check By: _____
J. LAF Sanitization Disinfectant used: <u>X%</u> H₂O₂	Room #: _____ LAF #: _____ Batch #: _____ Exp. Date: _____ By: _____
K. Mix for 5 minutes then pull the following samples during transfer of antigen into carboys. Label each carboy with Batch #, date, and volume. Retention = 4x3mL- freeze in lab (-70°C) Bacteria = 1x3mL- store in cooler (4°C) until delivered to QC	By: _____ Check By: _____

Procedure	Data Entries
L. Seal carboys with white tape around lid and transfer them to the freezer (refer to SOP XX-XXX-X). First at $\leq -20^{\circ}\text{C}$ then at $\leq -40^{\circ}\text{C}$.	Freezer #: _____ Temp: _____ $^{\circ}\text{C}$ Date: _____ Time: _____ AM / PM By: _____ Check By: _____
	Freezer #: _____ Temp: _____ $^{\circ}\text{C}$ Date: _____ Time: _____ AM / PM By: _____ Check By: _____
VII. SUB Filter Decontamination and Integrity Testing	
A. Decontamination	Date: _____ Autoclave # _____ Cycle # _____ By: _____
B. Filter Integrity Testing Refer to SOP XX-XXX-X – Operation of the Millipore Integritest 5 Automatic Filter Integrity Test Instrument.	Date: _____ By: _____
C. Refer to GEN XXX for testing procedure. Attach Integrity Test Results to this manufacturing direction for each filter tested.	By: _____
VIII. Pre-Inactivation	

Procedure	Data Entries
A. Harvest carboys should be pulled from freezer with ample time to thaw and warm to 3x°C.	Date: _____ Time: _____AM / PM By: _____
B. Pre-warm Sterile WFI for BEA/BEI cyclization to 3x°C the night before inactivation.	Date: _____ Time: _____AM / PM By: _____
IX. Cyclization of BEA to BEI (Binary ethylamine) <i>NOTE: BEI is a potential mutagen, extreme caution should be taken during the preparation of BEI solution to avoid inhalation or skin contact. The whole procedure should be conducted in a ventilation hood and/or using PAPR.</i>	
A. Calculate Volume of BEI needed Use this table to calculate and record the components used.	Antigen Volume _____mL (Step VI.H.) (Convert to mL) ÷ 9 Volume of BEI = _____mL Actual Vol. Formulated = _____mL

Component	Manufacturer / Batch #	Exp. Date	Amount/ mL	Volume to Add	By	Check By	SAP
BEA			0.0205g	g			
NaOH Pellets			0.007g	g			
Sterile WFI	In House		1mL	L			

Procedure	Data Entries
B. Measure components as determined by table. Add NaOH pellets to 3L of WFI and allow to completely dissolve and warm to 3x°C ($\pm 1^\circ\text{C}$).	
C. Combine BEA powder with approximately xx% of total WFI, and then add NaOH solution. Adjust to final formulation volume with WFI.	Cyclization Start Time: _____ AM / PM Temp: _____ °C Date: _____ By: _____ Check By: _____
D. Stir solution at 3x°C ($\pm 1^\circ\text{C}$) for 1 hour to complete cyclization.	Cyclization Stop Time: _____ AM / PM Temp: _____ °C By: _____ Check By: _____
E. Record pH of solution after cyclization and adjust pH to 7.x – 7.x. NOTE: If pH has not dropped to $\leq 8.x$, the solution should be neutralized with sodium thiosulfate and discarded. NOTE: Use BEI solution within x hours of cyclization stop time.	Post-Cyclization pH: _____ Final pH: _____ HCl Batch #: _____ Expiration Date: _____ Amount Added: _____ mL NaOH Batch #: _____ Expiration Date: _____ Amount Added: _____ mL By: _____ Check By: _____

Procedure	Data Entries
X. Inactivation A. Follow SOP XX-XXX-X to install the SUM bag inside the reservoir tank. Verify that the Bag is not wrinkled at the bottom before inserting the RTD sensor. Alternatively, a sterilized tank can be used for inactivation.	Date: _____ SUM #: _____ Bag Serial #: _____ OR Tank #: _____ Sterilization Exp. Date: _____ By: _____ Check By: _____
B. Make connections to the inactivation bag/tank and ensure that swing jaws are off and the appropriate lines are open.	
C. Verify SUM scale is reading Zero before transferring the virus fluid. Alternatively, a sterilized tank can be placed on a floor scale to determine weight. Floor scale should be calibrated and zero'd prior to use.	By: _____ Check By: _____
D. Transfer virus fluids pre-warmed to 3x°C from the carboys to the inactivation bag/tank. Record the final volume of the transferred antigen.	Volume of Virus Fluid = _____ mL By: _____ Check By: _____
E. Follow SOP XX-XXX-X, if applicable, to sign in to the SUM controller. Install the mixer, set the following parameters: Temperature = XX°C Agitator = XXX rpm Alternatively, the sterilized tank with antigen can be stored in an available incubator at XX°C. Ensure that tank is mixing.	Date: _____ By: _____ Check By: _____

Procedure	Data Entries
F. Record pH of virus fluids and adjust to $7.x \pm 0.3$ if necessary.	Initial pH: _____ Final pH: _____ HCl Batch #: _____ Expiration Date: _____ Amount Added: _____ mL NaOH Batch #: _____ Expiration Date: _____ Amount Added: _____ mL By: _____ Check By: _____
G. Add BEI solution to viral fluids, filtering it through a $0.xx\mu\text{m}$ filter. Ensure SUM bag/tank is mixing properly during transfer.	Volume of BEI Added: _____ mL New Antigen Volume: _____ mL (scale value) By: _____ Check By: _____
H. Immediately transfer virus fluids from the inactivation bag to another SUM bag/tank. Follow the procedure described in step X – E to set parameters, limits, and enable these alarms. Temperature = $3x^{\circ}\text{C}$ Agitator = $1xx\text{ rpm}$ Hi Hi alarm = $3x^{\circ}\text{C}$ Lo Lo alarm = $3x^{\circ}\text{C}$ Hi Hi alarm = $1xx\text{ rpm}$ Lo Lo alarm = $9x\text{ rpm}$ If using a tank, place tank with antigen/BEI solution in an incubator at $3x^{\circ}\text{C}$. Ensure that tank is mixing.	Date: _____ SUM #: _____ Bag Serial #: _____ OR Tank #: _____ Sterilization Exp.Date: _____ By: _____ Check By: _____

Procedure	Data Entries
I. Virus fluid should be mixed slowly at $3x^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for ~xx hr.	Inactivation Start Time: _____ AM / PM Temp: _____ $^{\circ}\text{C}$ Date: _____ Verify Mixing By: _____ Check By: _____ Inactivation Stop Time: _____ AM / PM Temp: _____ $^{\circ}\text{C}$ Date: _____ Verify Mixing By: _____ Check By: _____
J. Neutralized the remaining BEI using 3M NaThiosulfate. Run 3M NaThiosulfate into the first SUM bag/tank through the drain tube before placing the bag in a yellow barrel and label it as Hazardous waste, if applicable.	3M NaThiosulfate Batch #: _____ Exp. Date: _____ Volume of BEI Formulated: _____ mL Volume of BEI Added: _____ mL Volume of BEI Discarded: _____ mL Vol. of 3M NaThio added: _____ mL By: _____
XI. Neutralization	Date: _____

Procedure				Data Entries		
Neutralization Component Table						
Components	Manufacturer	Batch Number	Exp. Date	By	Check By	SAP
3M NaThio						
Gentamycin						
Thimerosal						
A. At the end of inactivation, add 1mL of 3M Sodium Thiosulfate per XXXmL of inactivated virus fluids to yield a final concentration of 30mM. Record appropriate information in Neutralization Component Table. Antigen Volume from Step X.G. (Scale Value)				Antigen Volume: _____ ÷ 100 Vol. of 3M NaThio: = _____ Antigen Vol.: _____ Vol. of 3M NaThio: + _____ New Antigen Vol: = _____		
B. Allow Antigen to neutralize for 1 to 4 hours mixing slowly at XXXrpm.				Start Time: _____ AM / PM Temperature: _____ °C Stop Time: _____ AM / PM Temperature: _____ °C		
C. When Neutralization is complete, turn off the parameters and disable the alarms, then transfer antigen into a dispense tank, if using a SUM bag. If using a tank, the tank can be sent for dispense following neutralization. After the transfer, place the SUM bag in a yellow barrel and label it as Hazardous waste, if applicable.				Dispense Tank #: _____ Sterilization Exp. Date: _____ By: _____ Check By: _____		

Procedure	Data Entries
D. Add XXX mg/mL Gentamycin solution for a final concentration of 0.000XmL / mL of neutralized fluid. Record appropriate information in Neutralization Component Table.	New Antigen Vol.: _____ mL $\times$ <u>0.000X</u> Gentamycin Vol.: = _____ mL By: _____ Check By: _____
E. Pull the following post-inactivation samples before continuing. NOTE: Samples must specify that they do not contain Thimerosal. Retention (before Thimerosal) = 5 x 10mL (store at -70°C) Inactivation = 4 x 10mL (deliver to QC 4°C cooler)	By: _____
F. Add 10% Thimerosal for a final concentration of 1:10,000. Record appropriate information in Neutralization Component Table. Final antigen Volume = New antigen vol. + Gent. Vol. – 90mL of sample	New Antigen Vol.: _____ mL Gentamycin Vol.: + _____ mL - <u>90mL samples</u> Final Antigen Vol.: = _____ mL $\div$ <u>1000</u> Thimerosal Vol.: = _____ mL
G. Record pH of antigen and adjust to $7.x \pm 0.1$ if necessary.	Initial pH: _____ Final pH: _____ HCl Batch #: _____ Amount Added: _____ mL NaOH Batch #: _____ Amount Added: _____ mL By: _____ Check By: _____

Procedure		Data Entries																																				
H. Pull the following post-inactivation samples before continuing. NOTE: Samples must specify that they contain thimerosal. Retention(after Thimerosal) = 5 x 10mL store at -70°C Densitometry = 4 x 10mL – Deliver to QC 4°C cooler. Sterility = 2 x 10mL – Deliver to QC 4°C cooler.		By: _____																																				
I. Dispense antigen into pools and store at ≤ -20°C. Label each pool with material #, batch #, pool #, pool volume, and dispensing date.		By: _____ Check By: _____																																				
Bulk Disposition <table border="1"> <thead> <tr> <th>Pool Number</th> <th>Volume / Pool</th> <th>Total Volume</th> <th>Freezer #</th> <th>Temp</th> <th>By</th> <th>Check By</th> </tr> </thead> <tbody> <tr> <td></td> <td>mL</td> <td>mL</td> <td></td> <td>-20°C</td> <td></td> <td></td> </tr> <tr> <td></td> <td>mL</td> <td>mL</td> <td></td> <td>-20°C</td> <td></td> <td></td> </tr> <tr> <td></td> <td>mL</td> <td>mL</td> <td></td> <td>-20°C</td> <td></td> <td></td> </tr> <tr> <td colspan="2">Total Volume</td> <td>mL</td> <td colspan="4"></td> </tr> </tbody> </table>				Pool Number	Volume / Pool	Total Volume	Freezer #	Temp	By	Check By		mL	mL		-20°C				mL	mL		-20°C				mL	mL		-20°C			Total Volume		mL				
Pool Number	Volume / Pool	Total Volume	Freezer #	Temp	By	Check By																																
	mL	mL		-20°C																																		
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Total Volume		mL																																				
J. Record the batch number and the amount of bottles used for dispense.		Bottles Batch #: _____ Amount Used: _____ ea By: _____ Check By: _____																																				
K. Verify all pools for label accuracy before sending them to -20°C storage freezer.		By: _____																																				

Procedure	Data Entries
L. Submit the appropriate samples to QC for testing.	Date Submitted: _____ By: _____
M. SAP and Excel transactions.	SAP By / Date: _____ / _____ Excel By / Date: _____ / _____

EQUIPMENT STERILIZATION RECORD

Attach all sterilization charts, integrity test verifications, process charts, formulation records etc. that support the documentation of the processes performed within this document.

Place stickers from one piece of equipment or glassware from each sterilization run used.

Attaching sterilization stickers verifies that all packed sterilized equipment has been checked for integrity and found acceptable for use.

MEDIA PREPARATION

Date: _____ By: _____

INOCULATION

Date: _____ By: _____

HARVEST:

Date: _____ By: _____

INACTIVATION:

Date: _____ By: _____

NEUTRALIZATION / DISPENSE

Date: _____ By: _____

COMMENTS

Comments: _____

Rev (dd mmm yy)

BP MFG US 2000L SUB

Reviewed By: _____

Date: _____